THE SOURCE OF A MORPHISM OF FINITE TYPE OVER A LOCALLY NOETHERIAN STACK IS LOCALLY NOETHERIAN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The presentation of a morphism of algebraic stacks through a commutative square of schemes with smooth surjective vertical and horizontal morphisms. Used at the start of the counted proof (source lines 3532-3541). Reduces the assertion to a statement about schemes.
- The statement that the source of a morphism of finite type of schemes whose target is locally Noetherian is locally Noetherian. Invoked at the end of the counted proof (source lines 3543-3545) as a cross-chapter pointer. Provides the conclusion for the presenting scheme.

Source reference. Original reference morphisms-lemma-finite-type-noetherian: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2240–2240. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-finite-type-noetherian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. If f is locally of finite type and $\mathcal{Y}$ is locally Noetherian, then $\mathcal{X}$ is locally Noetherian.*

Proof. Let

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{Y} \end{array}$$

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be a commutative diagram where U, V are schemes, $V \to \mathcal{Y}$ is surjective and smooth, and $U \to V \times_{\mathcal{Y}} \mathcal{X}$ is surjective and smooth. Then $U \to V$ is locally of finite type. If $\mathcal{Y}$ is locally Noetherian, then V is locally Noetherian. By Morphisms, Lemma morphisms-lemma-finite-type-noetherian we see that U is locally Noetherian, which means that $\mathcal{X}$ is locally Noetherian. $\square$